

AN APPLICATION OF TWO NOVEL MODELS FOR DISCRETE-VALUED TIME SERIES TO BIRTHS DATA

Iain L. MacDonald: Department of Business Science, University of Cape Town, Rondebosch, 7700, South Africa

Key words: BIC; hidden Markov time series models; logistic autoregressive models.

Summary: This paper applies several novel models for discrete-valued time series to a series previously analyzed by Haines, Munoz and Van Gelderen (1989, *J. Appl. Statist.*, 16, 55-67), in their study of births at Edendale Hospital in Natal, South Africa. The two classes of model considered are linear logistic autoregressive models and hidden Markov models. Both types of model are explicitly discrete-valued and provide information not available from the usual Gaussian ARIMA models. The Bayesian information criterion is used to compare the models proposed.

1. INTRODUCTION

Haines, Munoz and Van Gelderen (1989) have described the fitting of Gaussian ARIMA models to various discrete-valued time series related to births occurring during a 16-year period at Edendale Hospital in Natal, South Africa. The data include (inter alia) monthly totals of mothers delivered and of deliveries by various methods at the Obstetrics Unit of that hospital in the period from February 1970 to January 1986 inclusive.

One of the series considered by Haines et al., to which they fitted two models, was the number of deliveries by Caesarean section. From their models they drew the conclusions (in respect of this particular series) that there is a clear dependence of present on past observations, and that there is

STMA 1993 SUBJECT CLASSIFICATION: 14:040(12:070)

a clear linear upward trend. In this paper we describe the fitting of models of two novel types to this series. These models are rather different from those fitted by Haines et al. in that they are based on the binomial distribution and are discrete-valued: their models, being based on the normal distribution, are continuous-valued. The discrete-valued models make it possible to model the proportion (as opposed to the number) of Caesareans performed in each month. The two new classes of model proposed differ from each other in that one is 'observation-driven' and the other 'parameter-driven', in the sense in which Cox (1981) uses those terms. The most important conclusion that will be drawn from the discrete-valued models, and one which the Gaussian ARIMA models did not provide, is that there is a strong upward time trend in the proportion of the deliveries that are by Caesarean section.

2. THE DATA

The data used by Haines et al. (1989) included monthly counts of the following, for the period from February 1970 to January 1986 inclusive: mothers delivered, Caesarean sections performed, breech births, forceps deliveries and deliveries by vacuum extraction. Although sixteen years of data were available, Haines et al. considered only the final eight years' observations when modelling any series other than the total deliveries. This was because, for some of the series they modelled, the model structure was found not to be stable over the full sixteen-year period. In this study we have also modelled only the last eight years' observations.

In table 1 we present all the data used in this paper: mothers delivered, Caesareans, breech births, forceps deliveries and vacuum extraction deliveries. The data are given for the final nine years rather than the final eight, because certain of the logistic autoregressive models described in section 4 incorporate a lag of up to twelve months.

3. ARIMA MODELS FOR THE MONTHLY NUMBER OF CAESAREANS

The two models which Haines et al. fitted to the time series of Caesareans performed, and which they found to fit very well, may be described as follows. Let Z_t denote the number of Caesareans in month t (February 1978 being month 1), and let the process $\{a_t\}$ be Gaussian white noise, i.e. uncorrelated random shocks distributed normally with zero mean and common variance σ_a^2 . The first model fitted is the ARIMA(0,1,2) model with constant term:

$$(1) \quad \nabla Z_t = \mu + a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2}.$$

The maximum likelihood estimates of the parameters, with associated standard errors, are: $\hat{\mu} = 1,02 \pm 0,39$, $\hat{\theta}_1 = 0,443 \pm 0,097$, $\hat{\theta}_2 = 0,393 \pm 0,097$ and $\hat{\sigma}_a^2 = 449,25$. The second model is an AR(1) with linear trend:

$$(2) \quad Z_t = \beta_0 + \beta_1 t + \phi Z_{t-1} + a_t,$$

with parameter estimates as follows: $\hat{\beta}_0 = 120,2 \pm 8,2$, $\hat{\beta}_1 = 1,14 \pm 0,15$, $\hat{\phi} = 0,493 \pm 0,092$ and $\hat{\sigma}_a^2 = 426,52$.

It may be seen that both of these models provide support for the conclusion of Haines et al. that there is a dependence of present on past observations, and a linear upward trend. Furthermore, the models are nonseasonal: the Box-Jenkins methodology used found no seasonality in the Caesareans series. The X-11-ARIMA seasonal adjustment method employed in an earlier study (Munoz, Haines and Van Gelderen, 1987) did, however, find some evidence, albeit weak, of a seasonal pattern in the Caesareans series similar to a pattern that was observed in the 'total deliveries' series. (This latter series shows marked seasonality, with a peak in September, and in Haines et al. (1989) is modelled by the seasonal ARIMA model $(0, 1, 1) \times (0, 1, 1)_{12}$.)

It is of some interest to model the proportion, rather than the number,

of Caesareans in each month. It could be the case, for instance, that any trend, serial dependence or seasonality apparently present in the number of Caesareans is largely inherited from the total deliveries, and a constant proportion Caesarean is an adequate model. On the other hand, it could be the case that there is an upward trend in the proportion of the deliveries that are by Caesarean and this accounts at least partially for the upward trend in the number of Caesareans. In the next two sections we shall discuss two classes of models that condition on the total number of deliveries in each month and seek to describe the principal features of the proportion Caesarean.

4. LINEAR LOGISTIC AUTOREGRESSIVE MODELS

Let n_t denote the total number of deliveries in month t . A very general possible model for $\{Z_t\}$, the number of Caesareans, which allows for trend, dependence on previous observations and seasonality in the proportion Caesarean is as follows. Suppose that, conditional on the history $Z^{(t-1)} = \{Z_s : s \leq t-1\}$, Z_t is distributed binomially with parameters n_t and p_t , where for some positive integer q

$$(3) \quad \text{logit } p_t = \alpha_1 + \alpha_2 t + \beta_1 (Z_{t-1}/n_{t-1}) + \beta_2 (Z_{t-2}/n_{t-2}) + \dots \\ + \beta_q (Z_{t-q}/n_{t-q}) + \gamma_1 \sin(2\pi t/12) + \gamma_2 \cos(2\pi t/12).$$

This generalizes the model Cox (1981) describes as an 'observation-driven linear logistic autoregression', in that it incorporates trend and seasonality and is based on a binomial rather than a Bernoulli distribution. It is observation-driven in the sense that the distribution of the observation at a given time is specified in terms of the observations at earlier times. Clearly it is possible to add further terms to the above expression for $\text{logit } p_t$ to allow for the effect of any further covariates thought relevant, e.g. the number or proportion of deliveries by various instrumental techniques.

It does not seem possible to formulate a full maximum likelihood procedure to estimate the parameters $\alpha_1, \alpha_2, \beta_1, \dots, \beta_q, \gamma_1$ and γ_2 of the

model (3). It is, however, relatively easy to compute estimates of these parameters by using a program such as GLIM or GENSTAT to maximize a conditional likelihood. If for instance no observations earlier than Z_{t-1} appear in the model, the product

$$\prod_{t=1}^{96} \binom{n_t}{z_t} p_t^{z_t} (1-p_t)^{n_t-z_t}$$

is the likelihood of $\{Z_t : t = 1, \dots, 96\}$, conditional on Z_0 . Maximization thereof with respect to $\alpha_1, \alpha_2, \beta_1, \gamma_1$ and γ_2 yields estimates of these parameters, and can be accomplished simply by performing a logistic regression of Z_t on $t, Z_{t-1}/n_{t-1}, \sin(2\pi t/12)$ and $\cos(2\pi t/12)$.

In the search for a suitable model the following explanatory variables were considered: t (i.e. the time in months, with February 1978 as month 1), t^2 , the proportion Caesarean lagged one, two, three or twelve months, sinusoidal terms at the annual frequency (as in (3)), the calendar month, the proportion and number of deliveries by forceps or vacuum extraction, and the proportion and number of breech births. Model selection was performed by minimizing the Bayesian information criterion (BIC) proposed by Schwarz (1978) and used, for instance, by Raftery (1985). This is given by $-2l + k \log T$, where l is the maximized log likelihood of the model under discussion, k the number of parameters estimated and T the number of observations. GLIM, the program used, does not provide l , but does provide the (scaled) deviance, which is twice the difference between the maximum log likelihood achievable in a full model and that achieved in the model under investigation (McCullagh & Nelder, 1989, p. 33). Here the maximum log likelihood of a full model is $-321,039$, from which it follows that $-l = 321,039 + \frac{1}{2} \times \text{deviance}$. The best models found with between one and four explanatory variables (other than the constant term) are listed in table 2, as well as several other models that may be of interest. These other models are: the model with constant term only, the model with Z_{t-1}/n_{t-1} , the previous proportion Caesarean, as the only explanatory variable, and two models which replace

the *number* of forceps deliveries as covariate by the *proportion* of instrumental deliveries. The last two models were included because the proportion of instrumental deliveries may seem a more sensible explanatory variable than the one it replaces: it will be observed, however, that the models involving the number of forceps deliveries are preferred by BIC. (By instrumental deliveries we mean those which are either by forceps or by vacuum extraction.) BIC indicated that very little would be gained by inclusion of a fifth explanatory variable in the model.

The strongest conclusion we may draw from these models is that there is indeed a marked upward time trend in the proportion Caesarean. Secondly, there is positive dependence on the proportion Caesarean in the previous month. The negative association with the number (or proportion) of forceps deliveries is not surprising in view of the fact that delivery by Caesarean and by forceps are in some circumstances alternative techniques. As regards seasonality, the only possible seasonal pattern found in the proportion Caesarean is the positive 'October effect'. Among the calendar months only October stood out as having some explanatory power. As can be seen from table 2, the indicator variable specifying whether the month was October was included in the 'best' set of four explanatory variables. A referee has made the interesting suggestion that a possible reason for the October effect is that a relatively large number of Caesareans are performed on overdue mothers.

5. HIDDEN MARKOV MODELS

The main conclusion emerging from the models of the previous section is that there is a marked upward time trend in the proportion Caesarean. Another class of discrete-valued time series models, which also can account for time trend in that proportion in the presence of autocorrelation, are the hidden Markov models, which may be described briefly as follows. There is an underlying and unobserved sequence of states which follows a Markov chain with finite state-space, and the probability distribution of the observation at any time is determined by the current state of that Markov chain. Further-

more observations at different times are conditionally independent: that is, conditional on the Markov chain they are independent. Such models have long been used in speech processing: see for instance Levinson, Rabiner and Sondhi (1983) or Juang and Rabiner (1991). (It should be noted, however, that the hidden Markov models described in those papers do not assume that the Markov chain is stationary, whereas here we shall assume just that.) The name 'hidden Markov model' is apparently due to L.P. Neuwirth (Poritz, 1988). Zucchini and Guttorp (1991) recently applied hidden Markov models to the modelling of the wet-dry sequence at one or several rainfall stations – that is, they used them as multivariate binary time series models.

Hidden Markov models are an example of parameter-driven processes: conditional on an unobserved 'parameter process' the observations of a parameter-driven process are independent, with distribution determined by the current state of that parameter process. Cox and Snell (1989, p. 101) comment that parameter-driven processes 'tend to be difficult to handle as a basis for the analysis of data', but in fact hidden Markov models are an exception to this rule, and are surprisingly easy to handle. The likelihood of a sequence of observations assumed to come from a hidden Markov model can be computed and subjected to direct numerical maximization in order to estimate parameters.

The hidden Markov models we use in this application have two states and are defined as follows. Suppose $\{C_t\}$ is a stationary homogeneous Markov chain on state space $\{1, 2\}$, with transition probability matrix

$$\Gamma = \begin{bmatrix} 1 - \gamma_1 & \gamma_1 \\ \gamma_2 & 1 - \gamma_2 \end{bmatrix}.$$

Suppose also that, conditional on the Markov chain, Z_t has a binomial distribution with parameters n_t and p_t , where $C_t = i$. A model without time trend assumes that p_1 and p_2 are constants which, along with γ_1 and γ_2 , need to be estimated to determine the model completely. One

possible model which allows p_i to depend on t has $\log p_i = \alpha_i + \beta t$ and requires the estimation of five parameters. A more general model yet, with six parameters, has $\log p_i = \alpha_i + \beta_i t$.

It can be shown (for all three models) that the unconditional likelihood (L_{96}) of the observations $Z_1, \dots, Z_{96}$ is given by

$$(4) \quad L_{96} = \delta \lambda_1(z_1) \Gamma \lambda_2(z_2) \Gamma \dots \Gamma \lambda_{96}(z_{96}) 1,$$

where $\delta = \frac{1}{\gamma_1 + \gamma_2} (\gamma_2 \quad \gamma_1)$ is the stationary distribution of the Markov chain, $1 = (1 \quad 1)'$, and:

$$\lambda_i(z) = \begin{bmatrix} \begin{bmatrix} n_i \\ z \end{bmatrix} p_1^z (1 - p_1)^{n_i - z} & 0 \\ 0 & \begin{bmatrix} n_i \\ z \end{bmatrix} p_2^z (1 - p_2)^{n_i - z} \end{bmatrix}.$$

Provided that the computations are suitably scaled in order to prevent underflow, it is fairly straightforward to evaluate this likelihood, or rather its logarithm. To be more specific, the expression (4) for the likelihood is of the form

$$a \left[\prod_{t=2}^T B_t \right] 1,$$

where $a = \delta \lambda_1(z_1)$ and $B_t = \Gamma \lambda_t(z_t)$, and may in principle be evaluated merely by postmultiplying a successively by $B_2, \dots, B_T$ and 1. Scaling is necessary, however, because the elements of $u = a \prod_{t=2}^T B_t$ are typically very small, and underflow even for quite small r . What was done in this work was to scale the vector u at each stage of the multiplication in such a way that the average element is 1, i.e. to divide u by $m^{-1} \sum_{i=1}^m u_i$, and to accumulate the logarithms of these scale factors. Once the scaled likelihood has been computed thus, the sum of the logs of the scale factors can be added to the log of the scaled likelihood, and the result is the log-likelihood.

Maximization of the likelihood, subject to the bounds of 0 and 1 on the probabilities involved, can be performed by a simple derivative-free unconstrained optimization algorithm: the Nelder-Mead simplex method, as implemented by Press et al. (1986), was used here. It should be pointed out, however, that maximum-likelihood fitting of hidden Markov models with three or more states requires maximization subject to generalized upper bound constraints on the transition probabilities, and cannot be handled by an unconstrained optimization algorithm.

The three hidden Markov models fitted, along with the associated log-likelihood and BIC values, appear in table 3. (In that table, t as usual denotes the time in months and February 1978 is month 1.) It may be seen that, of the three models, that with a single time-trend parameter and a total of five parameters achieves the smallest BIC value. That model can be described as consisting of a Markov chain with two fairly persistent states, along with their associated time-dependent probabilities of delivery being by Caesarean, the (upward) time trend being the same for the two states. State 1 is rather more likely than state 2, because the stationary distribution of the Markov chain is (0,618 0,382), and has a lower probability of delivery being by Caesarean. It may or may not be possible to interpret the states as (for instance) non-busy and busy periods in the Obstetrics Unit of the hospital, but without further information, e.g. on staffing levels, such an interpretation would be speculative. Even without such interpretation, however, hidden Markov models can be useful descriptive tools: 'empirical' models rather than 'substantive' ones (Cox, 1990).

6. DISCUSSION

The difference in likelihood between the hidden Markov models with and without time trend is convincing evidence of an upward trend in the proportion Caesarean, and confirms the main conclusion drawn from the models of section 4. Although Haines et al. concluded that there is an

upward trend in the number of Caesareans, it does not seem possible to draw any conclusion about the proportion Caesarean from their models, or from any other ARIMA models.

It is of interest also to compare the fit of the five-parameter hidden Markov model to the data with that of the logistic autoregressive models. It is notable that the hidden Markov model produces a lower value of BIC (827,52) than does the 'best' logistic autoregressive model with four explanatory variables (840,50), and makes use of less information. It does not use z_0 , nor does it use information on forceps deliveries or the calendar month.

It seems therefore that hidden Markov models have considerable potential as simple yet flexible models for examining dependence on covariates (such as time) in the presence of autocorrelation. Details of the theory of these models, and various extensions thereof, are presented in the thesis of the author (MacDonald, 1992). One aspect worth highlighting here, however, is the ease with which forecasts can be produced by hidden Markov models. For instance, if one wishes to use such a model to forecast the proportion Caesarean at time 97 for a given number of deliveries, what is needed is the one-step-ahead forecast distribution of Z_{97} , i.e. the distribution of Z_{97} conditional on $Z_1, \dots, Z_{96}$. This is given by the likelihood of $Z_1, \dots, Z_{97}$ divided by that of $Z_1, \dots, Z_{96}$:

$$\frac{L_{97}}{L_{96}} = \frac{\delta\lambda_1(z_1)\Gamma\lambda_2(z_2) \dots \Gamma\lambda_{97}(z_{97})1}{\delta\lambda_1(z_1)\Gamma\lambda_2(z_2) \dots \Gamma\lambda_{96}(z_{96})1}.$$

More generally, it can be shown that the k -step-ahead forecast distribution, i.e. the conditional probability that $Z_{96+k} = z$, is:

$$\frac{\delta\lambda_1(z_1)\Gamma\lambda_2(z_2) \dots \Gamma\lambda_{96}(z_{96})\Gamma^k\lambda_{96+k}(z)1}{\delta\lambda_1(z_1)\Gamma\lambda_2(z_2) \dots \Gamma\lambda_{96}(z_{96})1}.$$

ACKNOWLEDGEMENTS

Prof. Walter Zucchini is thanked for suggesting this work, and for many helpful discussions. Dr Linda Haines is thanked for making the data available. The author is grateful to the referees for their constructive comments, which did much to improve the paper.

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Table 1. The data

Key: (1) mothers delivered
 (2) Caesarean sections performed
 (3) breech births
 (4) forceps deliveries
 (5) vacuum extraction deliveries.

month	(1)	(2)	(3)	(4)	(5)	month	(1)	(2)	(3)	(4)	(5)
2/77	553	87	6	35	35	2/80	523	130	27	38	13
3/77	612	96	14	29	27	3/80	540	127	15	36	14
4/77	635	104	9	30	41	4/80	533	156	21	32	7
5/77	671	110	18	58	51	5/80	534	149	26	26	17
6/77	692	107	39	82	50	6/80	553	144	35	29	20
7/77	672	124	24	86	49	7/80	636	150	11	35	22
8/77	690	122	31	70	31	8/80	604	158	15	38	26
9/77	767	161	17	56	46	9/80	670	168	25	32	34
10/77	627	130	5	48	35	10/80	627	152	27	26	34
11/77	586	117	0	46	44	11/80	532	101	25	15	16
12/77	691	138	17	32	48	12/80	565	132	26	19	18
1/78	645	131	12	32	38	1/81	562	124	14	52	27
2/78	594	141	34	24	29	2/81	544	127	9	46	36
3/78	594	117	10	40	43	3/81	602	154	7	44	32
4/78	607	134	30	42	20	4/81	582	126	35	39	58
5/78	658	122	10	61	14	5/81	615	127	16	42	51
6/78	559	118	12	34	19	6/81	606	152	15	34	47
7/78	614	125	21	44	21	7/81	643	157	25	23	32
8/78	673	146	18	56	20	8/81	672	145	15	32	43
9/78	753	168	21	62	23	9/81	768	191	20	37	41
10/78	656	161	25	47	16	10/81	679	178	15	48	37
11/78	657	141	24	37	11	11/81	629	147	25	37	35
12/78	648	167	31	44	20	12/81	719	188	15	45	36
1/79	558	124	26	37	23	1/82	685	177	6	57	36
2/79	571	152	16	58	15	2/82	591	154	10	37	21
3/79	626	155	13	77	4	3/82	651	145	10	18	26
4/79	571	102	27	81	28	4/82	622	172	14	23	32
5/79	618	167	30	67	32	5/82	703	171	10	28	34
6/79	584	147	23	58	16	6/82	661	168	12	46	23
7/79	605	133	24	58	14	7/82	714	210	10	26	56
8/79	659	142	11	64	10	8/82	776	227	14	28	58
9/79	687	135	36	81	15	9/82	869	239	6	28	78
10/79	615	157	32	55	3	10/82	742	202	8	36	51
11/79	592	126	25	51	10	11/82	699	198	24	19	46
12/79	552	120	22	41	11	12/82	710	183	21	16	72
1/80	598	130	31	35	18	1/83	710	197	19	19	53

month	(1)	(2)	(3)	(4)	(5)
2/83	711	185	14	18	43
3/83	760	234	17	23	55
4/83	764	225	11	13	55
5/83	741	249	19	15	45
6/83	673	196	19	19	53
7/83	743	178	8	35	60
8/83	775	222	7	23	44
9/83	896	221	18	28	85
10/83	727	192	20	27	45
11/83	649	173	17	9	28
12/83	722	185	14	12	50
1/84	680	196	10	13	51
2/84	716	208	7	19	68
3/84	755	225	15	14	48
4/84	736	212	10	9	46
5/84	740	198	19	9	40
6/84	780	204	24	3	43
7/84	743	193	16	15	38
8/84	864	219	22	23	54
9/84	921	249	17	18	60
10/84	820	234	6	19	28
11/84	777	207	15	30	18
12/84	847	210	23	42	17
1/85	817	207	18	32	45
2/85	773	192	27	17	60
3/85	883	209	26	27	66
4/85	777	205	13	16	53
5/85	826	201	14	24	51
6/85	784	223	19	19	54
7/85	824	222	25	20	61
8/85	853	258	19	8	64
9/85	970	287	14	26	46
10/85	831	261	17	20	35
11/85	744	221	18	14	41
12/85	817	207	15	14	49
1/86	780	186	18	12	37

Table 2. Models fitted by GLIM to the proportion Caesarean (logit link).

explanatory variables	coefficients	deviance	BIC
constant t (time in months)	-1,253 0,003439	208,92	860,13
constant t previous proportion Caesarean	-1,594 0,002372 1,554	191,70	847,47
constant t previous proportion Caesarean no. forceps deliveries in month t	-1,445 0,001536 1,409 -0,002208	183,63	843,97
constant t previous proportion Caesarean no. forceps deliveries in month t October indicator	-1,446 0,001422 1,431 -0,002393 0,08962	175,60	840,50
constant	-1,073	324,05	970,69
constant previous proportion Caesarean	-1,813 2,899	224,89	876,10
constant t previous proportion Caesarean proportion instrumental deliveries in month t	-1,505 0,002060 1,561 -0,7507	188,32	848,66
constant t previous proportion Caesarean proportion instrumental deliveries in month t October indicator	-1,528 0,002056 1,590 -0,6654 0,07721	182,36	847,26

Table 3. The three 2-state hidden Markov models fitted to the proportion Caesarean

logit p_i	$-l$	BIC	γ_1	γ_2	α_1	α_2	β_1	β_2
α_i	420,322	858,90	,059	,086	-1,184	-,9601	-	-
$\alpha_i + \beta t$	402,349	827,52	,162	,262	-1,317	-1,140	,003297	
$\alpha_i + \beta_i t$	402,314	832,01	,161	,257	-1,315	-1,150	,003249	,003467